General Mathematics — Exercise Sheet 35

Forced oscillations and Fourier coefficients

Variant: Statements

For a first pass through the lecture's main ideas, begin with the beats and Fourier exercises (numbers 2 and 4). The first problem is a cross-course extension combining the fundamental theorem, integration by parts and second-order uniqueness. The starred sixth problem is a further investigation; the complete solution remains available if you want to follow its argument.

Exercise 1 (Variation of parameters). For a continuous real function f on an open interval containing zero, set

$$y(t) = -\cos t \int_0^t f(s)\sin s\,\mathrm{d}s + \sin t \int_0^t f(s)\cos s\,\mathrm{d}s.$$

Prove directly that this function is the unique solution of $y'' + y = f(t)$ with $y(0) = y'(0) = 0$. This construction is called *variation of parameters*. Apply it to $f(t) = te^t$, evaluating both integrals. Use Lecture 26's fundamental theorem, Lecture 8's product rule, Lecture 23's integration by parts and Lecture 34's homogeneous uniqueness.

Exercise 2 (Nearby frequencies and beats). Let $\omega_0 > 0$, $\omega \geq 0$ and $\omega \neq \omega_0$. Solve $y'' + \omega_0^2 y = F\cos(\omega t)$ from rest, where $F > 0$. For $\omega_0 = 10$, $\omega = 11$ and $F = 1$, express the result as a product of sines. Identify the period of its nonnegative amplitude envelope, meaning the absolute value of the factor multiplying the faster sine. This slow variation is called *beating*. For each fixed t, find the limit of the general solution as $\omega \to \omega_0$. Use Lectures 34–35's forcing formulas, Lecture 2's trigonometric identities and Lecture 7's derivative definition for the frequency limit.

Exercise 3 (Circuit current, width and quality factor). Assume the LRC circuit model

$$LQ'' + RQ' + \frac{Q}{C} = V\cos(\omega t), \qquad L, C, R, V > 0, \quad \omega \geq 0,$$

with current $I = Q'$. Find the amplitude of the steady current and the frequency where it is largest. For $\omega > 0$ write its amplitude using $L\omega - 1/(C\omega)$. Assume instantaneous resistor power is RI^2, so a sinusoidal current of amplitude I_a has mean power $RI_a^2/2$ over a period. Find the two positive frequencies ω_- and ω_+ at which this mean power is half its maximum. Compute their difference, and the *quality factor* $Q = \omega_0/(\omega_+ - \omega_-)$, where $\omega_0 = 1/\sqrt{LC}$. Use Lecture 35's response amplitude and Lecture 8's differentiation rule.

Exercise 4 (The quadratic Fourier series and the Basel sum). Let f be the 2π-periodic extension of $f(x) = x^2$ on $[-\pi, \pi]$. Compute its Fourier coefficients, state the series sum at every real x, and use $x = \pi$ to evaluate $\sum_{n=1}^{\infty} 1/n^2$. Also prove uniform convergence of this Fourier series on $\mathbb{R}$. Use Lecture 26's integration by parts, Lecture 35's coefficient formulas and cited convergence theorem, and Lectures 29–30's p-series and majorant test. The sum identification inherits the cited theorem's [C] status.

Exercise 5 (A general period and a half-range cosine series). Let $L > 0$. Derive the real Fourier coefficient formulas for period $2L$ from Lecture 35 by the substitution $x = L\theta/\pi$. For $g(x) = x$ on $[0, L]$, take the even extension to $[-L, L]$ and then extend periodically. This defines a *half-range cosine expansion* of g: the Fourier series of its even periodic extension. Compute that expansion, including its endpoint values. Use Lecture 26's substitution and parts formulas and Lecture 35's cited convergence theorem. The sum identification has status [C].

Exercise 6 (A square wave and persistent overshoot ⋆). Let f be 2π-periodic, equal to -1 for $-\pi < x < 0$, to 1 for $0 < x < \pi$, and to zero at integer multiples of π. Compute its coefficients and state the Fourier limit at every point. For the partial sums with $N + 1$ odd harmonics, put

$$T_N(x) = \frac{4}{\pi} \sum_{j=0}^{N} \frac{\sin((2j+1)x)}{2j+1}, \qquad N \geq 0.$$

Show that the first positive local maximum is at $x_N = \pi/(2N + 2)$, and prove

$$\lim_{N\to\infty} T_N(x_N) = \frac{2}{\pi} \int_0^{\pi} \frac{\sin u}{u}\, \mathrm{d}u > 1.$$

The integrand at zero is defined as 1. This persistent overshoot near the jump is called the *Gibbs phenomenon*. Use Lectures 5–6's sine limit, Lectures 26 and 30's integration results, Lecture 35's cited pointwise theorem, and Lecture 11's finite Taylor formula. For the strict inequality you may use the given school bounds $3 < \pi < 22/7$.

Exercise 7 (Parseval as a stated tool). For a real 2π-periodic piecewise C^1 function, use the following *Parseval identity* as a given cited result [C]:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} f(x)^2\, \mathrm{d}x = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

This is the real-coefficient form of J. Lebl's *Basic Analysis II*, version 6.3 (15 May 2026), Theorem 11.8.13, p. 204, author's book. The complex coefficients obey Lecture 35's relations. Apply the identity to the 2π-periodic extension of $|x|$ to find $\sum_{n \text{ odd}} 1/n^4$ and then $\sum_{n=1}^{\infty} 1/n^4$. Use the previous half-range computation at $L = \pi$ and Lecture 29's p-series test and Lecture 3's sequence-limit laws. The derived values inherit [C].

Exercise 8 (Recovering oscillator parameters from a periodic response). A damped oscillator satisfies

$$y'' + 2\gamma y' + \omega_0^2 y = \cos t + \sin 2t, \qquad \gamma > 0, \quad \omega_0 > 0.$$

Its measured periodic response is $y(t) = (3\cos t + 2\sin t)/13 - \cos 2t/4$. Determine γ and ω_0, and check that both observed frequency components are consistent with the same parameters. Use Lecture 35's orthogonality to compare finite trigonometric coefficients and Lecture 8's derivative rules. The question asks for exact parameters in the stated model.